Module-4 Database

1) What is RDBMS?

RDBMS stands for Relational Database Management System where rational means the tables are related (joined) based on common fields.
RDBMS is the basis for SQL, and for all modern database systems like MS SQL Server, IBM DB2, Oracle, MySQL, and Microsoft Access.

2) What is SQL?

SQL is Structured Query Language, which is a computer language for storing, manipulating and retrieving data stored in relational database.
SQL is the standard language for Relation Database System.
All relational database management systems like MySQL, MS Access, Oracle, Informix and SQL Server use SQL as standard database language.

3) Write SQL Commands

DDL – Data Defination Language
DCL – Data Control Language
DML – Data Manipulation Language
DQL – Data Query Language

4) What is join?

JOIN is used to combine rows from two or more tables based on a related column between them.

5) Write type of joins.

INNER JOIN: returns rows when there is a match in both tables.
LEFT JOIN: returns all rows from the left table, even if there are no matches in the right table.
RIGHT JOIN: returns all rows from the right table, even if there are no matches in the left table.
FULL JOIN: returns rows when there is a match in one of the tables.

6) How Many constraint and describes it self?

Not Null- Ensures a column cannot have a NULL (empty) value.

Unique key- Ensures all values in a column are distinct from one another.

Primary key**-** A combination of NOT NULL and UNIQUE. It uniquely identifies each row in a table.

Foregin key- Prevents invalid data from being entered into a column, as it must match a value in another table's primary key

7) Difference between RDBMS vs DBMS

DBMS (Database Management System) manages data as files or hierarchical structures, usually for single users, while RDBMS (Relational DBMS) is an advanced version storing data in tables (rows/columns) with relationships.

8) What is an SQL alias?

An SQL alias is a temporary name assigned to a table or a column for the duration of a single query.

9) Write a query to create the table in Structured Query Language.

```
CREATE TABLE table_name (
    column1 datatype constraints,
    column2 datatype constraints,
    ...
);
```

10) Write a query to insert data into table

```
INSERT INTO table_name (column1, column2, column3)
VALUES (value1, value2, value3);
```

11) Write a query to update data into table with validations.

```
UPDATE Employees
SET Salary = Salary * 1.10
WHERE PerformanceRating = 'Excellent';
```

12) Write a query to delete data from table with validations.

```
DELETE FROM Customers
WHERE CustomerName = 'Alfreds Futterkiste'
    AND Country = 'Germany';
```

13) Write a query to insert new column in existing table.

```
ALTER TABLE table_name
ADD column_name data_type [constraint];
```

14) Write a query to drop table and database.

```
DROP TABLE table_name;
```

15) Write a query to find max and min value from table.

```
SELECT MIN(column_name) AS min value, MAX(column_name) AS max_value
FROM table_name;
```

16) Create two tables named Seller and Product apply foreign key in product table Fetch data from both table using different joins.

```
CREATE TBLE Seller (

    SellerID INT PRIMARY KEY,
    SellerName VARCHAR(100) NOT NULL,
    City VARCHAR(50)
);
```

17) What is API Testing?

API testing is a software testing practice that focuses on verifying the functionality, reliability, performance, and security of Application Programming Interfaces (APIs).

18) Types of API Testing

Open APIs: These types of APIs are publicly available to use like OAuth APIs from Google. It has also not given any restriction to use them. So, they are also known as Public APIs.
Partner APIs: Specific rights or licenses to access this type of API because they are not available to the public.
Internal APIs: Internal or private. These APIs are developed by companies to use in their internal systems. It helps you to enhance the productivity of your teams.

19) What is Responsive Testing?

In responsive testing we have to check layout/ui, breckpoints, navigation,images are properly set or not, font, scroll and etc.

20) Which types of tools are available for Responsive Testing?

LT Browser
Lembda Testing
Google Resizer
I am responsive
Pixel tuner

21) What is the full form of .ipa, .apk?

iOS App Store Package, Android Package Kit

22) How to create step for to open the developer option mode ON?

Go to the about phone then after click on build number 7 times and click on developer options – then after click on the usb debugging and final step is allow usb debugging.